# Ohm's Law and Combination of Resistors

**ENGS121 Electricity & Magnetism Lab**

**18th September, 2025**

## Introduction

Resistors are used to control the current and voltage distribution through the circuits. Those are made of a piece of pure carbon connected between two pins and are meant to have some resistance. The resistors are often connected in parallel or in series, and have their own laws depending on how they are connected. All of the rules are a consequence of the law of conservation of charge and Ohm's law. The goal of the experiment is to examine those and give an evaluation.

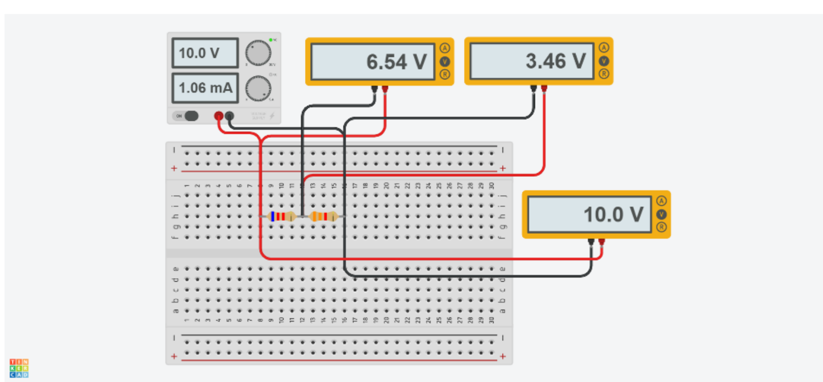
The resistors that are used in this experiment are meant to withstand no more than 0.25W of power. In order to calculate the maximum amount of allowed voltage for the resistors Ohm's Law can be used.

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{R}}, \boldsymbol{P} = \boldsymbol{U} * \boldsymbol{I}, \boldsymbol{P} = \boldsymbol{U} * \frac{\boldsymbol{U}}{\boldsymbol{R}} = \frac{\boldsymbol{U}^2}{\boldsymbol{R}} < \mathbf{0.25}\boldsymbol{W}$$

| Resistance (Ω) | 100 Ω | 500 Ω | 1 KΩ | 2 KΩ |
|---|---|---|---|---|
| Max. voltage (V) | 5 V | 11.2 V | 15.8 V | 22.4 V |

**Part 1: Voltage across the combination of resistors connected in series**

Our task here is to check the validity of the formula V = $V_1 + V_2$, across the combination of resistors connected in series.




*(Figure 1.1 setup of the experiment)*

## Instruments

| Name | Quantity | Description |
|---|---|---|
| Breadboard | 1 | Board meant for connecting electrical component |
| Jumper Wires | - | Male-Male interface |
| Resistors | - | Kit of resistors with various resistances |
| Voltmeter | 1 | Meant for measuring the voltage across the circuit |
| Power Supply 15V | 1pc | Imitates power through the circuit |

## Variables

| Variable | Type |
|---|---|
| $R_1$ | Parameter |
| $R_2$ | Parameter |
| $V_1$ | Measured |
| $V_2$ | Measured |
| $V_{All}$ | Control |
| $V_1 + V_2$ | Derived |

## Procedure

1) Connect the resistors in series
2) Measure the resistances by Ohmmeter
3) Measure the voltage across $R_1$, then across $R_2$, lastly through both

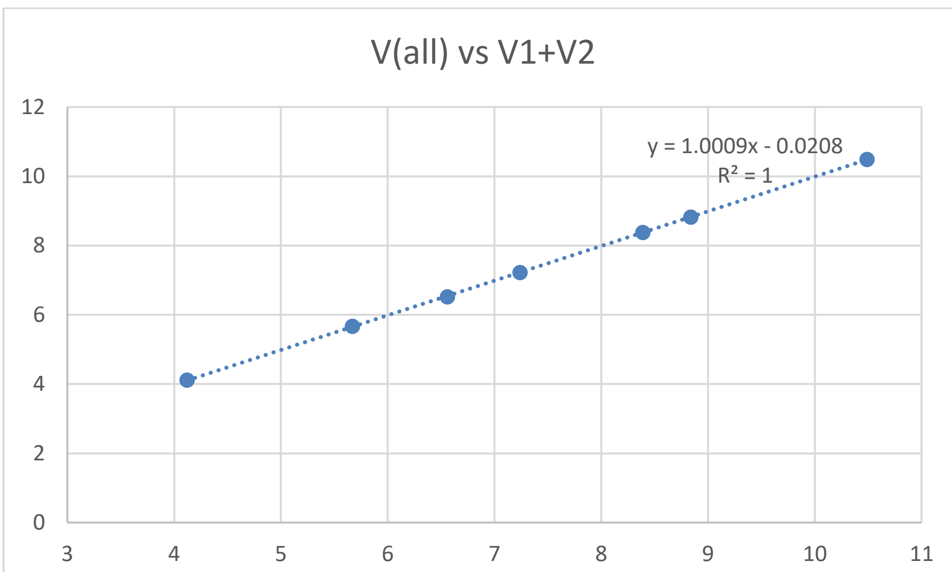
## Recorded Data

| R1 (fixed) | R2 (fixed) | V1 | V2 | V all | V1+V2 |
|---|---|---|---|---|---|
| 6.17 | 3.26 | 3.63 | 6.86 | 10.49 | 10.49 |
| | | 3.05 | 5.77 | 8.84 | 8.82 |
| | | 2.9 | 5.48 | 8.39 | 8.38 |
| | | 2.49 | 4.73 | 7.24 | 7.22 |
| | | 2.26 | 4.26 | 6.56 | 6.52 |
| | | 1.42 | 2.69 | 4.12 | 4.11 |
| | | 1.96 | 3.71 | 5.67 | 5.67 |

## Errors

| *Description* | *Type* | *Possible Solution* |
|---|---|---|
| Power Supply Output inaccuracy | Systematic | Don't trust the power supply information screen output, record the data measured by voltmeter instead |
| Breadboard Wiring Resistances | Systematic | These resistances in breadboard pins are often so small that it is not worth taking into account |
| Current Fluctuations | Random | Hold the multi meter pins straight until a solid reading appears |

## Plot

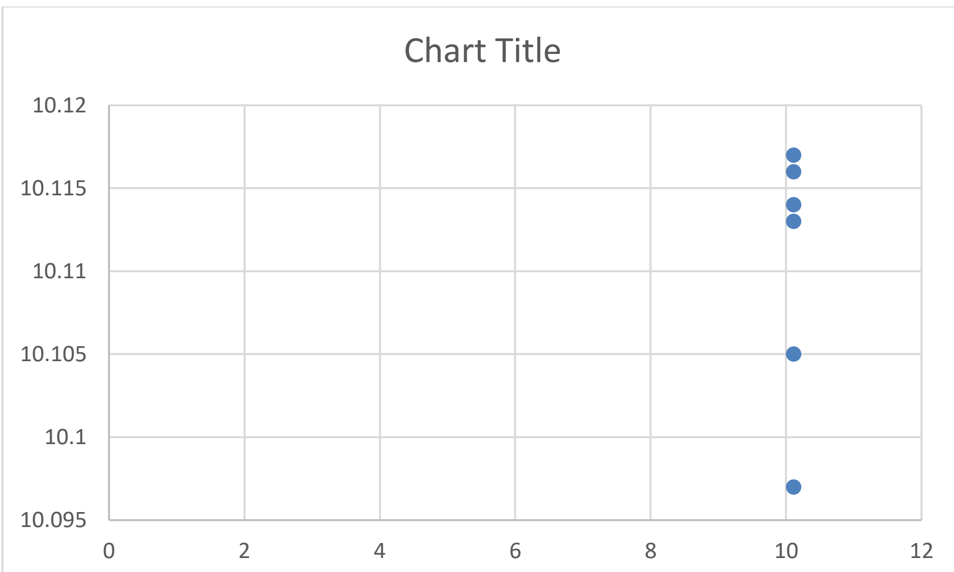



*(Figure 1.5 Chi Square for the first 210 rolls.)*

## Evaluation and Conclusion

As we can see there is a stronger linear regression between variables V all and V1 + V2. Moreover, the slope of the trendline appears to be 1, which shows a direct linkage between two sets of variables. Hence the statement V = V1 + V2 is reasonable to be true.

| R1 | R2 | V1 | V2 | Vall (fixed 10.11V) | V1 + V2 |
|---|---|---|---|---|---|
| 3.261 | 6.17 | 3.49 | 6.6 | 10.11 | 10.09 |
| 6.069 | 6.17 | 5.02 | 5.1 | 10.11 | 10.11 |
| 2.156 | 6.17 | 2.62 | 7.49 | 10.11 | 10.11 |
| 3.304 | 3.26 | 5.09 | 5.02 | 10.11 | 10.12 |
| 2.157 | 3.26 | 4.03 | 6.09 | 10.11 | 10.12 |
| 2.157 | 3.29 | 4.01 | 6.1 | 10.11 | 10.11 |




Since the plot of this case does not explain too much about the linkage between V all and V1 + V2, some other approach has to be used to evaluate the results. One way to do that is by computing the differences.

$\mu = 10.11\ V$

$\sigma = 0.01\ V$

The average value of V1 + V2 is exactly equal to the value of Vall

$\Delta V = 0$

The relative error is negligible, implies that **V1 + V2 => Vall is valid**.

**Part 2: Ohm's Law**

Check if the current through the circuit is proportional to across the resistor.

## Errors

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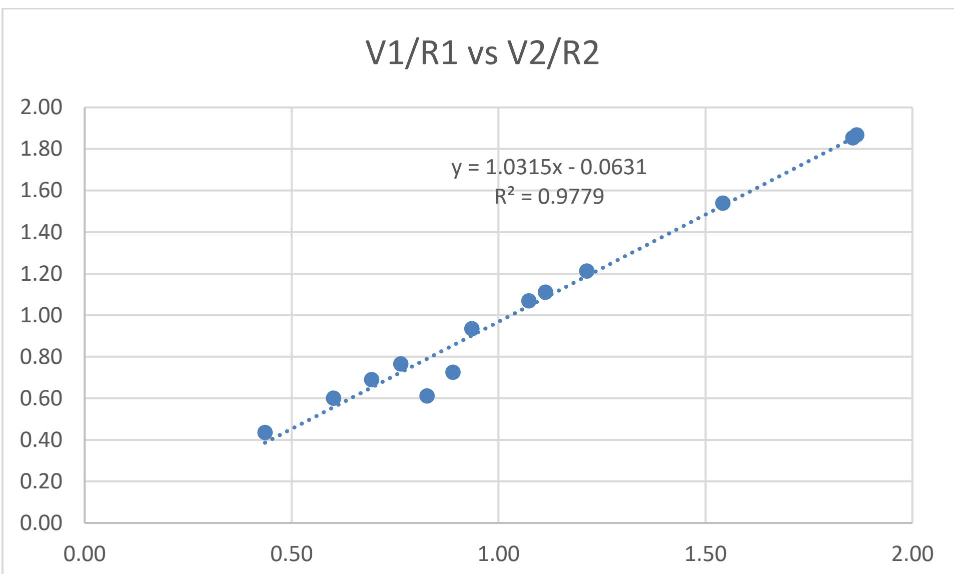
| Current Fluctuations | Random | Hold the multi meter pins straight until a solid reading appears |
| Resistor Factory Imperfections | Systematic | - |

| Variable | Type |
|---|---|
| $R_1$ | Parameter |
| $R_2$ | Parameter |
| $V_1$ | Measured |
| $V_2$ | Measured |
| $V_{All}$ | Control |
| V1/R1 | Derived |
| V2/R2 | Derived |

## Recorded Data

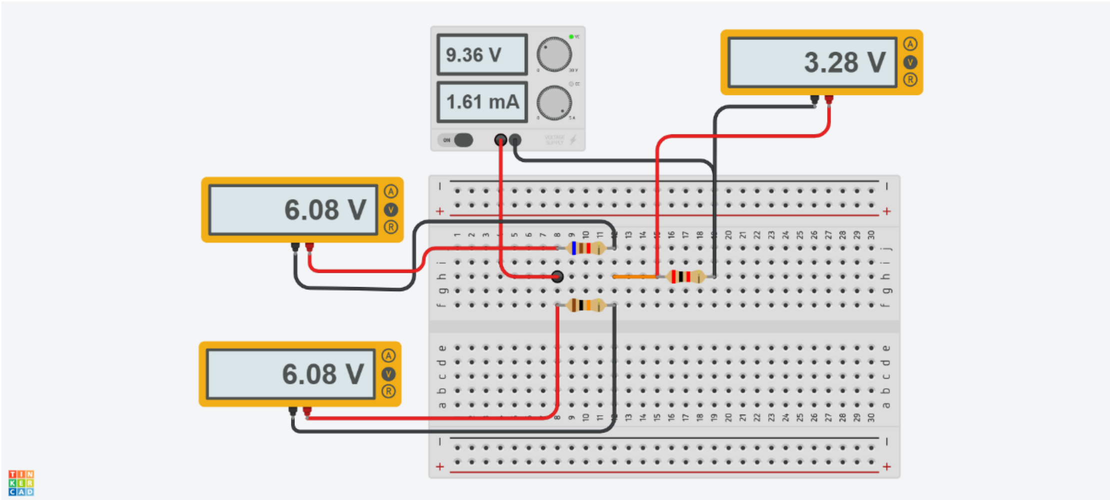
| R1 | R2 | V1 | V2 | Vall | V1/R1 | V2/R2 |
|---|---|---|---|---|---|---|
| 3.26 | 6.17 | 3.63 | 6.86 | 10.49 | 1.11 | 1.11 |
| 3.26 | 6.17 | 3.05 | 5.77 | 8.84 | 0.94 | 0.94 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 3.26 | 6.17 | 2.90 | 4.48 | 8.39 | 0.89 | 0.73 |
| 3.26 | 6.17 | 2.49 | 4.73 | 7.24 | 0.76 | 0.77 |
| 3.26 | 6.17 | 2.26 | 4.26 | 6.56 | 0.69 | 0.69 |
| 3.26 | 6.17 | 1.42 | 2.69 | 4.12 | 0.44 | 0.44 |
| 3.26 | 6.17 | 1.96 | 3.71 | 5.67 | 0.60 | 0.60 |
| 3.26 | 6.17 | 3.50 | 6.60 | 10.11 | 1.07 | 1.07 |
| 6.07 | 6.70 | 5.02 | 4.10 | 10.11 | 0.83 | 0.61 |
| 2.16 | 6.17 | 2.62 | 7.49 | 10.11 | 1.21 | 1.21 |
| 3.30 | 3.26 | 5.09 | 5.02 | 10.11 | 1.54 | 1.54 |
| 2.16 | 3.26 | 4.03 | 6.09 | 10.11 | 1.87 | 1.87 |
| 2.16 | 3.29 | 4.01 | 6.10 | 10.11 | 1.86 | 1.85 |




As we can see R^2 ≈ 0.978, which shows a very strong linear relationship, so **V1/R1≈V2/R2** indicating that the current is proportional to voltage as Ohm's law predicts. There are some minor deviations which are likely due to experimental errors, but overall, the validity of Ohm's Law is supported.

**Part 3: Testing the junction rule for parallel connection**




According to the conservation law of charge sum of the currents throught each resistor must be equal to the total current throught the combination.

| Variable | Type |
|---|---|
| $R_1$ | Control |
| $R_2$ | Control |
| $R_3$ | Control |
| $V_1$ | Measured |
| $V_2$ | Measured |
| $V_3$ | Measured |
| $V_{All}$ | Fixed |
| V1/R1 + V2/R2 | Derived |
| V3/R3 | Derived |

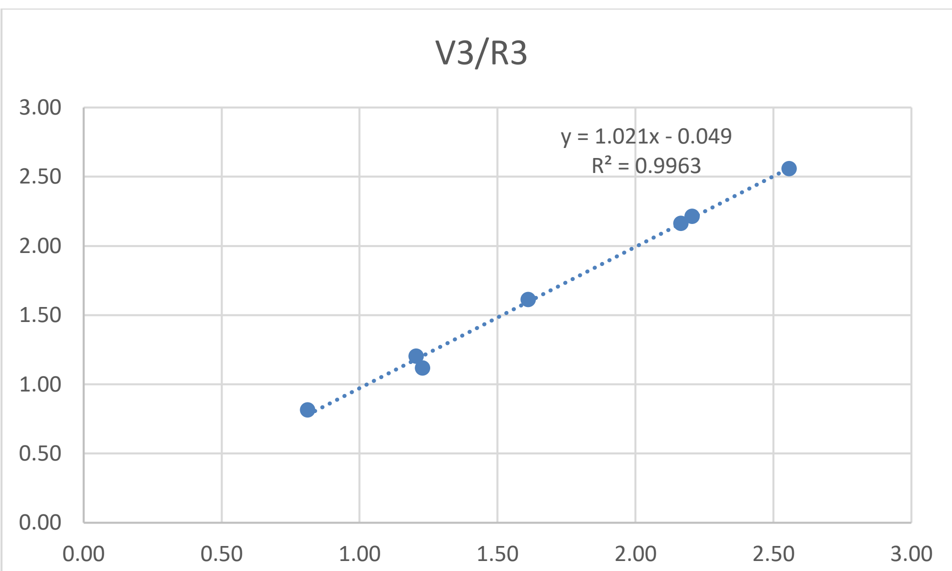
$$V_{All} = 9.36 \text{ V (fixed)}$$

## Errors

| *Description* | *Type* | *Possible Solution* |
|---|---|---|
| Power Supply Output inaccuracy | Systematic | Don’t trust the power supply information screen output, record the data measured by voltmeter instead |
| Breadboard Wiring Resistances | Systematic | These resistances in breadboard pins are often |

| | | |
|---|---|---|
| | | so small that it is not worth taking it into account |
| Current Fluctuations | Random | Hold the multi meter pins straight until a solid reading appears |
| Resistor Factory Imperfections | Systematic | - |

| R1 | R2 | R3 | V1 | V2 | V3 | $\frac{V_1}{R_1}+\frac{V_2}{R_2}$ | $\frac{V_3}{R_3}$ |
|---|---|---|---|---|---|---|---|
| 6.09 | 9.97 | 2.04 | 6.09 | 6.09 | 3.29 | 1.61 | 1.61 |
| 2.04 | 9.97 | 6.09 | 2.04 | 2.04 | 7.33 | 1.20 | 1.20 |
| 2.04 | 6.09 | 9.97 | 1.24 | 1.24 | 8.13 | 0.81 | 0.82 |
| 2.04 | 6.09 | 2.14 | 3.91 | 3.90 | 5.47 | 2.56 | 2.56 |
| 7.34 | 3.30 | 1.97 | 5.02 | 5.02 | 4.36 | 2.21 | 2.21 |
| 2.14 | 1.97 | 3.30 | 2.22 | 2.22 | 7.14 | 2.16 | 2.16 |
| 2.14 | 1.97 | 7.34 | 1.27 | 1.25 | 8.21 | 1.23 | 1.12 |




As we can see R^2 ≈ 0.996, which shows a very strong linear relationship, so $\frac{V_1}{R_1}+\frac{V_2}{R_2}\approx\frac{V_3}{R_3}$ indicating that the sum of the currents through R1 and R2 equals the current throught R3, supporting the validity of the junction rule.